GZW-1 Flame Arrester

General Inline Bidirectional Deflagration Flame Arrester

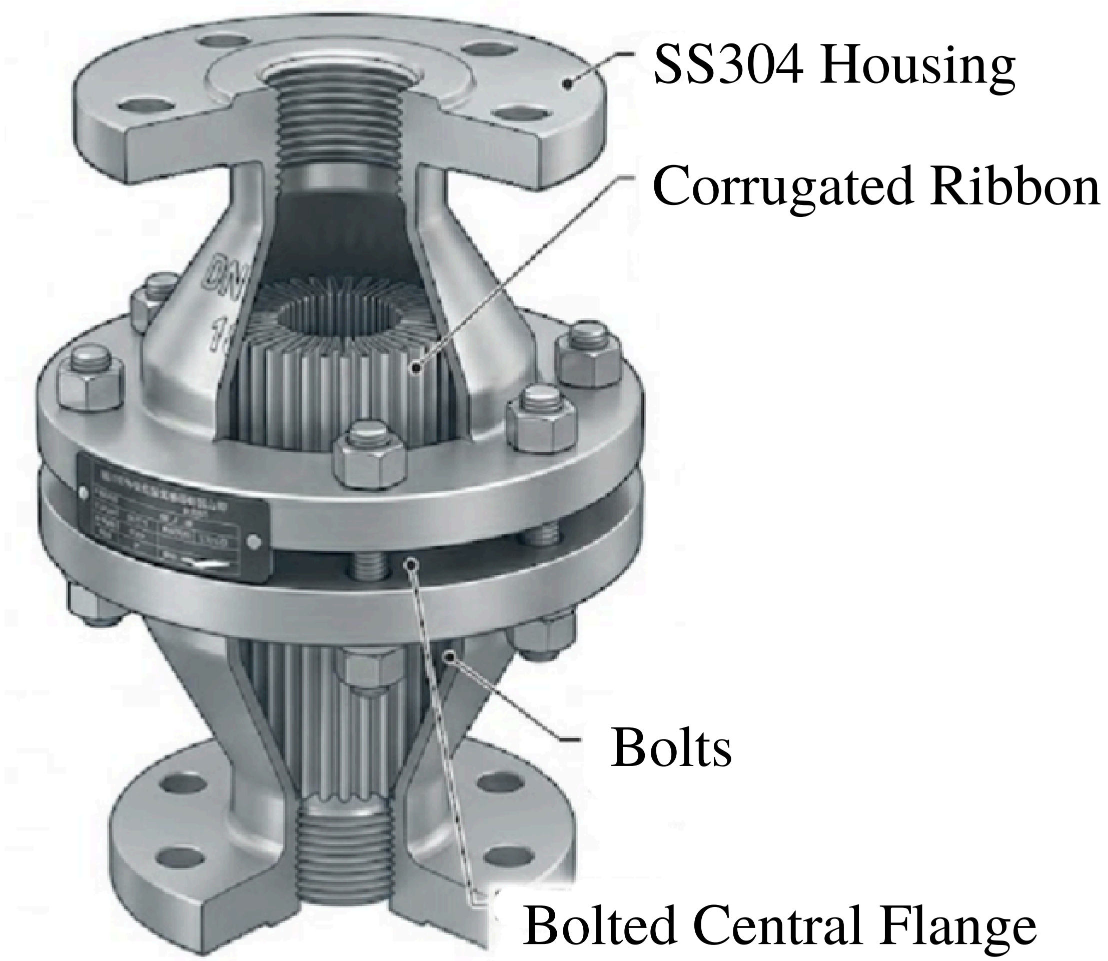
The hydraulic performance of the GZW-1 (DN40) has been validated through computational modeling based on established K-factors for crimped-ribbon elements (MESG <0.90mm). To maintain a robust Process Safety Margin, the following parameters were applied. While the product is commodified and many manufacturers design to this exact specification, the company used for this testing was Wenzhou Zhongwei Petrochemical Equipment Manufacturing Co. Ltd.:

- Conservatism Bias: A 25% Friction Penalty was integrated into standard international benchmarks. This overhead accounts for potential manufacturing tolerances, housing turbulence, and the specific fluid dynamics of localized vacuum environments.
- System Compatibility: This pressure drop represents a negligible load (< 0.5%) on the vacuum pump's suction capacity. Consequently, the inline arrester ensures zero risk of process choking or cavitation, maintaining full volumetric efficiency for the Stage III system.

- MESG: A Group IIA flame (gasoline) has an MESG of 0.93mm. Our Arrester is gapped at 0.90mm, providing a tighter physical barrier than required.
- Flame Endurance: Qualified for "short-time burning." The elements maintain structural integrity through 13 consecutive subsonic flame tests—more than double the ISO 16852 requirement of 5 tests.
- Shell Safety: Housing is rated for shell strength exceeding 1.5 * working pressure, verified via hydraulic qualification.

- **ATEX Compliance:** Technical Files deposited with Ente Certificazione Macchine (ECM) per ATEX Directive 2014/34/EU.
- **Manufacturing License:** TS2733J50-2029 (Special Equipment Production License).
- **Integrated Management:** Certified to ISO 9001:2015, ISO 14001:2015, and ISO 45001:2018.

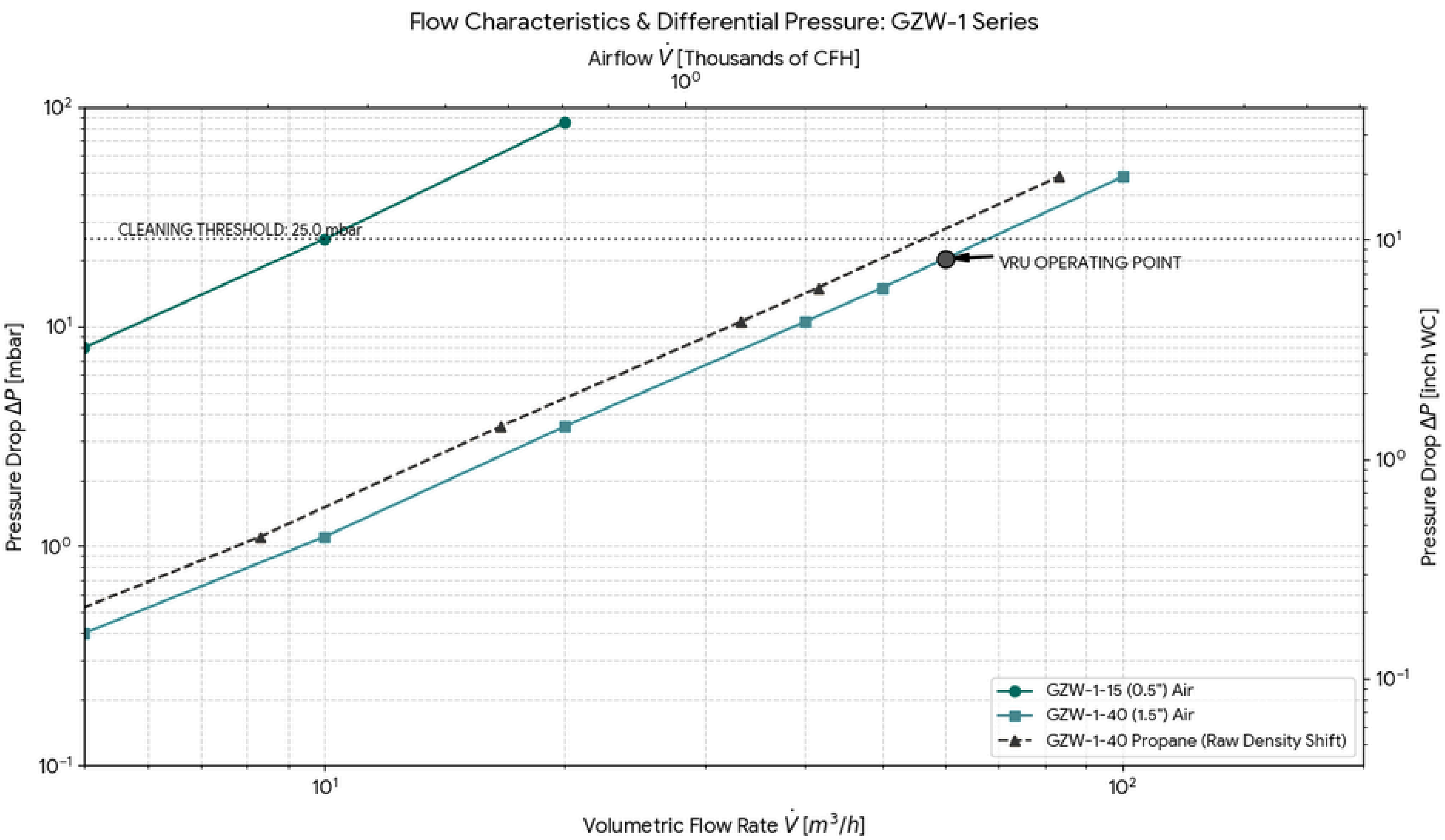
Feature	DN15 Specification	DN40 Specification
Nominal Diameter	DN15 (1/2" NPT)	DN40 (1-1/2" NPT)
Pressure Rating	PN16	PN16
Connection Type	Threaded (NPT)	Threaded (NPT)
Housing Material	304 Stainless Steel	304 Stainless Steel
Element Material	304 Stainless Steel	304 Stainless Steel
Gas Group (MESG)	IIA	IIA
Primary Standard	GB/T 13347	GB/T 13347



Performance Curves

Flow capacity V based on air of a density ρ = 1.29 kg/m³ or 1.87kg/m³ for propane at T = 273 K and atmospheric pressure p = 1.013 bar. For other gases the flow can be approximately calculated by:

$$\dot{V} = \dot{V}_b \cdot \sqrt{\frac{\rho_b}{1.29}} \quad or \quad \dot{V}_b = \dot{V} \cdot \sqrt{\frac{1.29}{\rho_b}}$$



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Performance Curves Addendum

The GZW-1 series arrester is installed in close proximity to the vapor recovery source, rendering the actual L_u/D ratio effectively zero.

However, to account for flow turbulence, pipe bends, and fittings within the line, we adhere to a more conservative safety factor: the L_u/D ratio shall be maintained at < 30, significantly exceeding the standard requirement of < 50 for Group IIA applications.

Explosion Group	Max L_u/D Ratio	DN40 (1.5") Pipe
Group IIA (Gasoline)	**50**	**2.0 Meters** (40mm × 50)

Pressure Analysis & System Performance
Total Suction Head: ~1,000 mbar System Margin: 100:1
Even at a peak flow of 60 m³/h, the pressure drop across the GZW-1 is so negligible compared to the total suction head that the pump will not "feel" the flame arrester in the line.

Flow Velocity Analysis: DN15 Utility Piping

- Flow Rate (Q): 1.0 m³/h (≈ 0.00028 m³/s)
- Pipe Area (A): A DN15 pipe with a 15 mm internal diameter has an area of approx. 0.000177 m²
- Velocity (v): Q / A ≈ 1.58 m/s

Performance Summary: The pump is rated for a nominal liquid flow of <1.0 m³/h, which corresponds to a stabilized line velocity of <1.5 m/s within the DN15 utility piping.

Performance Curves Addendum

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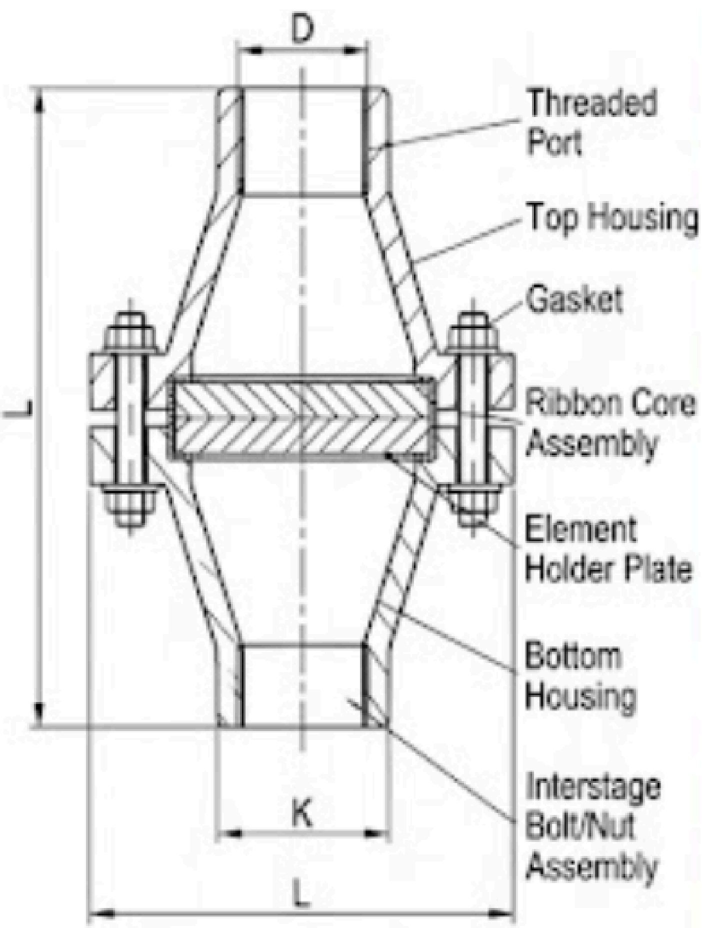
Performance Summary: The pump is rated for a nominal liquid flow of 1.0 m³/h, which corresponds to a stabilized line velocity of 1.58 m/s within the DN15 utility piping. This velocity is well within standard industrial limits for liquid transport, ensuring minimal friction loss and preventing cavitation at the pump inlet.

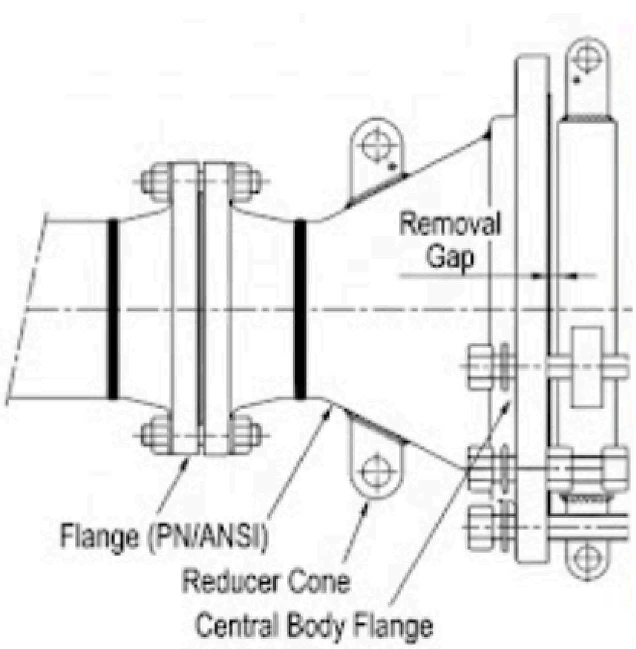
Technical Data Sheets



Dimension	DN15 (G ½)	DN40 (G 1½)
L (OAL, Flanged version)	280 / 11.02	350 / 13.78
L (OAL, Threaded	140 / 5.51	175 / 6.89
K (Outside Diameter)	100 / 3.94	137 / 5.39
D (Thread size)	G ½ / M15	G 1½ / M40
Removal Gap	15 / 0.59	20 / 0.79
SW (Flange Width)	10 / 0.39	14 / 0.55
Dimensions in mm / inch		

Parameter	Specification
MESG	≥ 1.14 mm
Explosion Group	IIA1

Connection Type	Standard
Threaded	ISO 228-1
Flanged	EN 1092-1, ASME



Parameter	DN15	DN40
P_{max} (bar / psi)	2.2 / 31.9	2.2 / 31.9
T_{max} (°C / °F)	≤ 60°C / 140°F	≤ 60°C / 140°F

Component	Material
Housing	Stainless Steel 316L
Gasket	PTFE (teflon)
FLAMEFILTER®	Stainless Steel 316L
Bolts	Stainless Steel A2-70
Special materials upon request.